

Solid State Physics

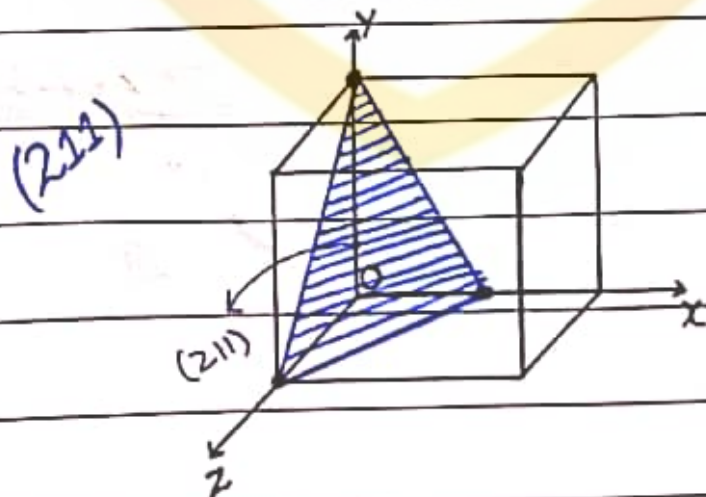
Remaining Past Paper

-: Question # 02:-

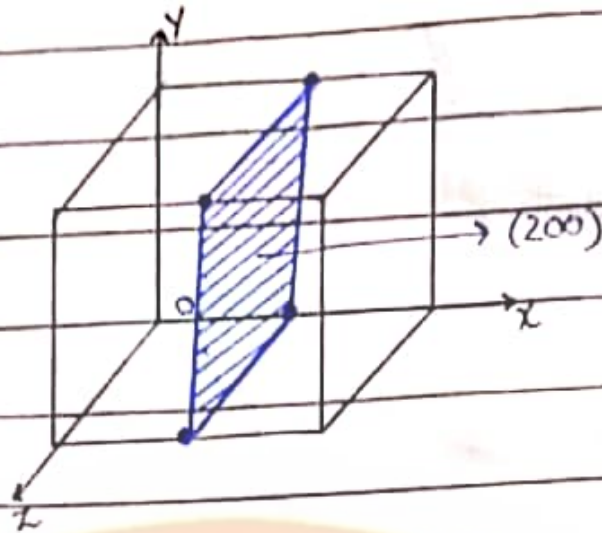
Give to the point answer/
short description of each
question.

(a)

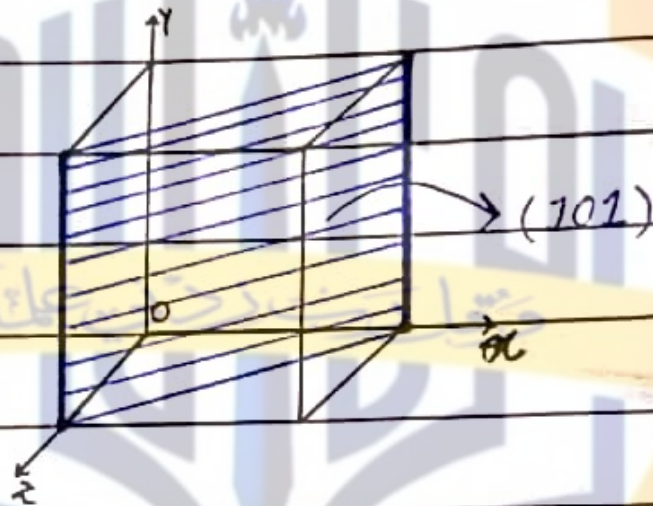
Draw (211), (200), (101) and
(111) crystallographic planes in
cubic unit cell.



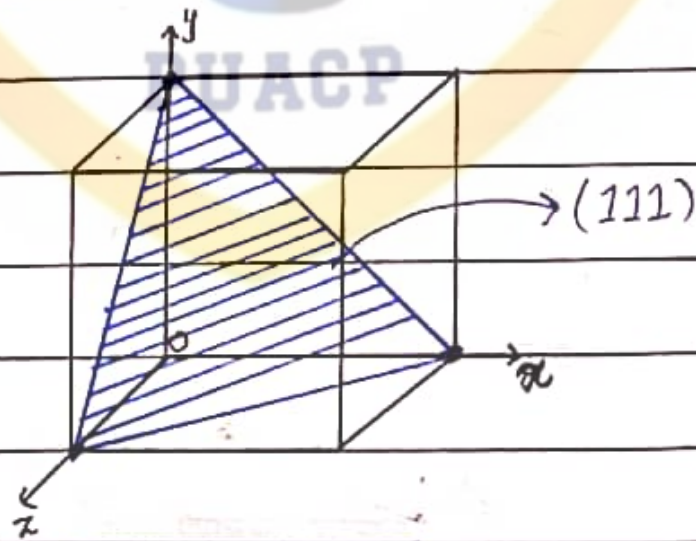
(200)



(101)



(111)



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(b)

Describe briefly what is meant by anharmonic crystal interactions-

Anharmonic crystal interactions refers to the interactions between atoms in a crystal lattice that cannot be fully described by harmonic approximations. In harmonic systems, the forces between atoms are assumed to be linearly related to atomic displacements, but in anharmonic systems, these forces have nonlinear dependencies. This nonlinearity leads to phenomena such as phonon-phonon interactions, which are crucial in understanding thermal properties of materials.

Example:

One common example is the creation of phonon-phonon interactions. These interactions involve the nonlinear coupling of vibrational

modes (phonons) in a crystal lattice. Phonons are quantized lattice vibrations, and in anharmonic systems, they can exchange energy with each other, leading to effects like thermal conductivity and lattice thermal expansion that cannot be explained by harmonic approximations alone.

(C)

Write down the steps to find Miller indices of a plane whose intercepts are 2, 1, 2.

Steps to find the Miller indices of planes are below.

Consider a general 3-D lattice.

→ Suppose we have to write the Miller indices of planes ABC.

→ Find the intercepts of planes on three axis in terms of lattice constant a, b, c

$2a, 1b, 2c$

→ Divide the intercepts as in ①
by their corresponding lattice constant
to get the numbers such as

$$\frac{2a}{a}, \frac{1b}{b}, \frac{2c}{c}$$
$$2, 1, 2$$

Taking reciprocal of above numbers

$$\frac{1}{2}, 1, \frac{1}{2}$$

as we need integer so we

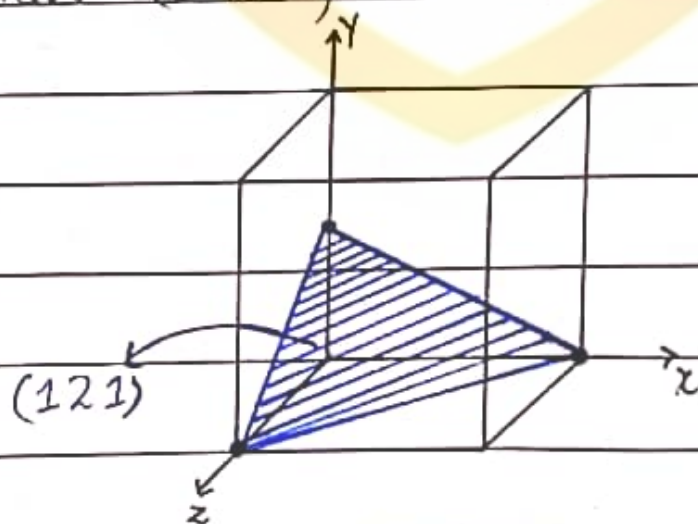
Multiply by all intercept with 2 and
get

$$1, 2, 1$$

Express the resulting integers in
lower case and enclose in round
brackets without any commas in
this case

$$(1\ 2\ 1)$$

Draw $(1\ 2\ 1)$



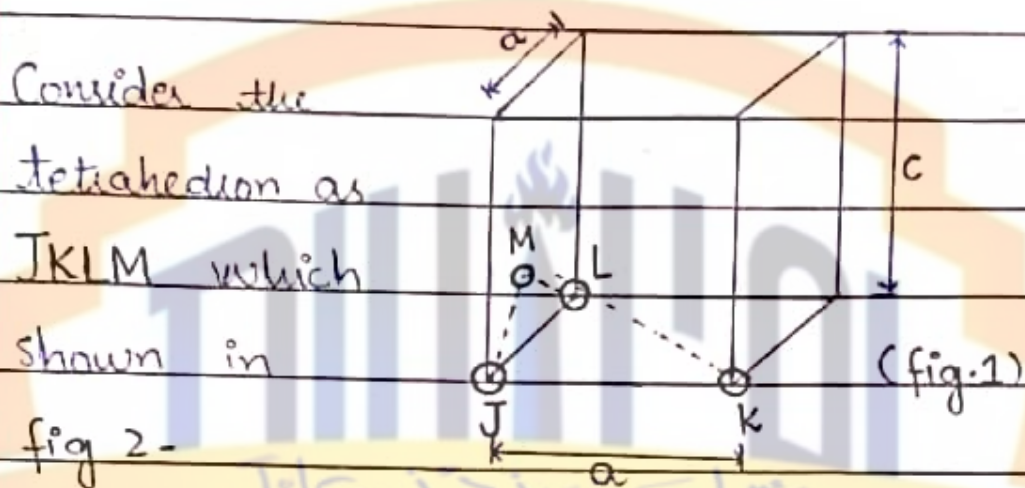
(d)
How an inert gas crystal holds together? Discuss the origin of Van der Waals interactions in crystals?

In Crystals, like those made of inert gases such as helium or neon, the atoms are held together by what we call van der Waals forces. These forces are a result of temporary dipoles that occur in the atoms due to fluctuations in electron distribution. These temporary dipoles create a weak attraction between neighbouring atoms, which helps hold the crystal structure together. While van der Waals forces are weaker compared to other types of bonds like covalent or ionic bonds, they are sufficient to maintain the stability of crystals at low temperatures and high pressures.

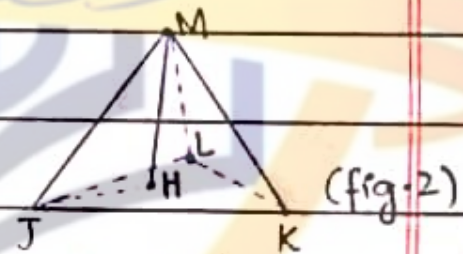
(e)

For hexagonal close-packed structure (HCP), show that $c/a = 1.633a$.

A sketch of one third of an HCP unit cell is shown in fig 1.



The atom at point M is mid way between top and bottom face of unit cell that is

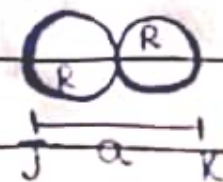


$$MH = \frac{c}{2} \text{ and}$$

Since atom at point J, K and M all touch one another.

$$JM = JK = 2R = a$$

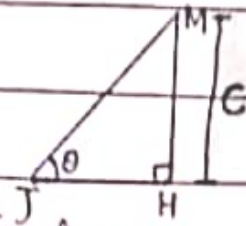
where R is atomic radius



Further make triangle JHM.

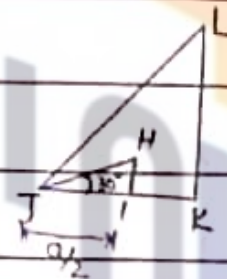
$$|JM|^2 = |JH|^2 + |MH|^2$$

$$a^2 = |JH|^2 + \left(\frac{c}{2}\right)^2 \rightarrow \text{①}$$



Now we can determine JH length by consideration JKL which is an equilateral triangle

$$\cos 30^\circ = \frac{a/2}{|JH|}$$



$$|JH| = \frac{a}{2} \cdot \frac{1}{\cos 30^\circ} = \frac{a \times \sqrt{3}}{2} \quad \because \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\overline{JH} = \frac{a}{\sqrt{3}}$$

Substitute the above value in eq ①

$$a^2 = \frac{a^2}{3} + \frac{c^2}{4}$$

$$\frac{c^2}{4} = a^2 - \frac{a^2}{3}$$

$$\frac{c^2}{4} = \frac{2a^2}{3}$$

$$\frac{c^2}{a^2} = \frac{8}{3}$$

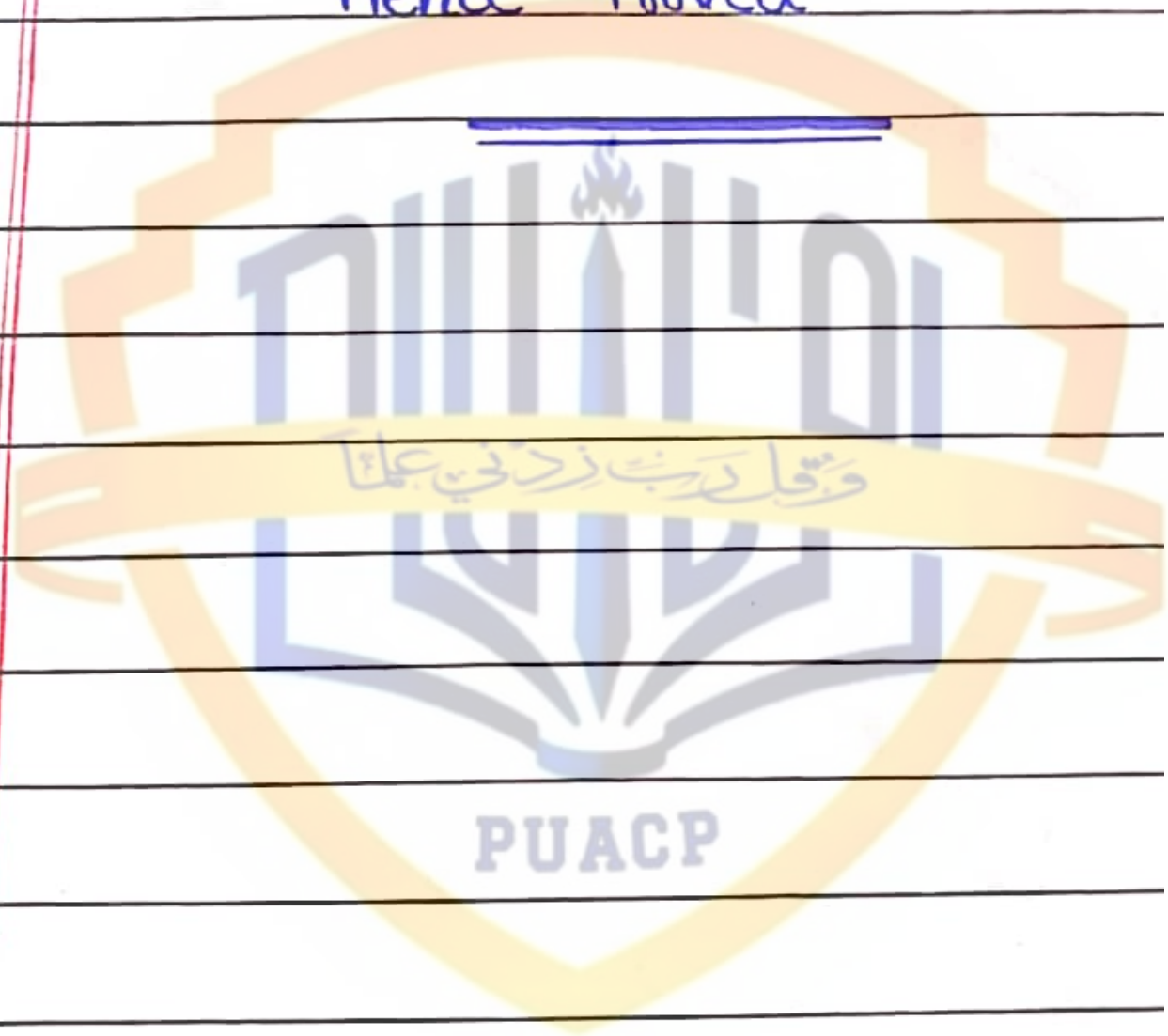
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$$\frac{c}{a} = \sqrt{\frac{8}{3}}$$

$$\frac{c}{a} = 1.633$$

Hence Proved



Question no 3 part (a)

Discuss vibrations in 1-dimensional di-atomic Linear Lattice.

Vibrations in 1-dimensional di-atomic Lattice:-

Consider a 1-D diatomic lattice with 2 different kinds of atoms (such as NaCl or CsCl) having masses M_1 & M_2 in one unit cell of lattice. Here

$M_2 > M_1$. Let us assume both masses are arranged alternatively along x-axis. Let the interatomic

separation b/w atoms in a lattice is constant and it is " a ".

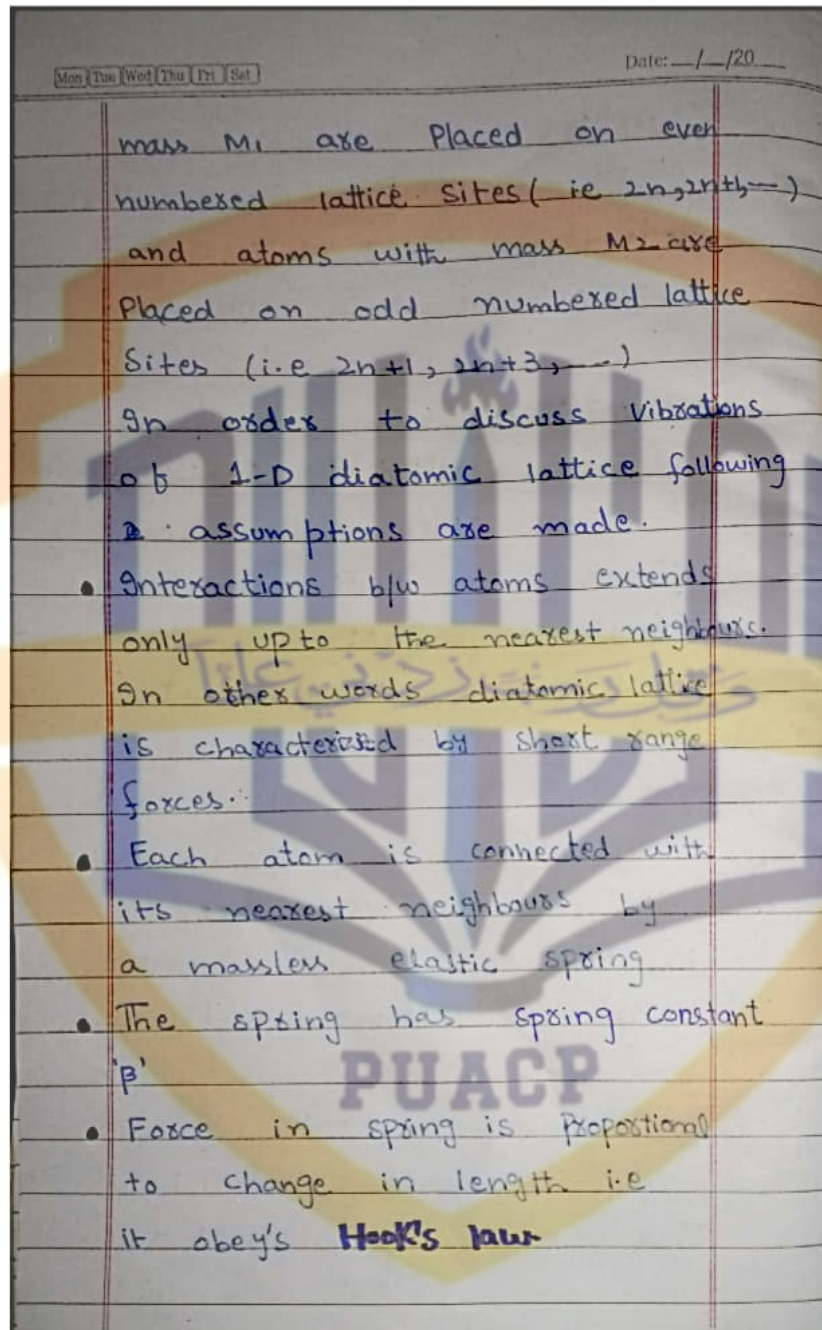
Then atoms are arranged in such a way that atoms with

← vibrations of 1 dimensional di-atomic...

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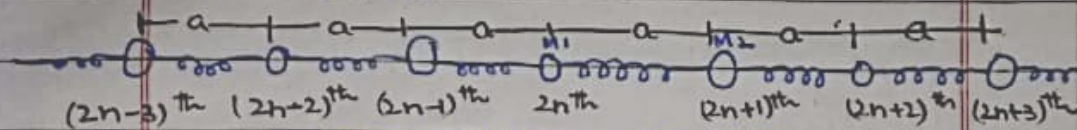
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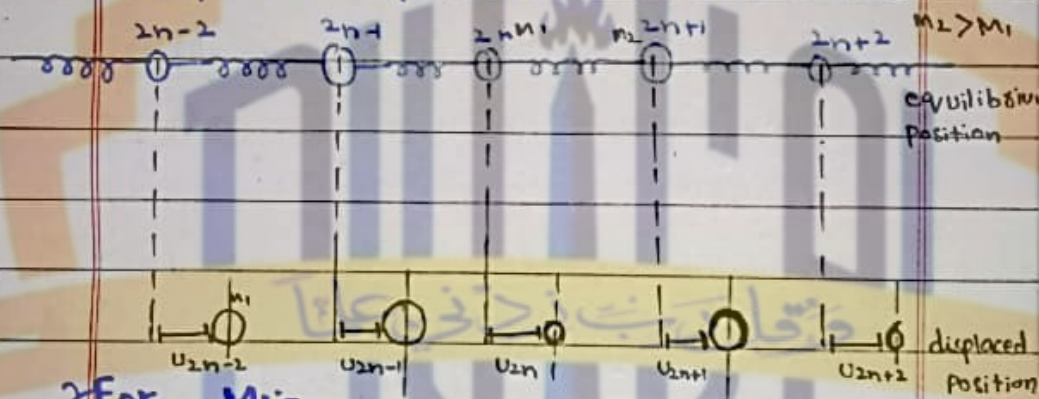
Chain of 1-D di-atomic lattice at equilibrium position is shown in figure.



■ equilibrium position.

All atoms are connected with each other by elastic spring so vibrations will travel throughout the medium.

Displaced position is shown as.



For m_1 :-

Force on $2n^{th}$ atom due of $(2n+1)^{th}$ atom:-

$$F_1 = -\beta U$$

$$\therefore F = -kx \quad \text{Hook's law}$$

$$F_1 = -\beta (U_{2n} - U_{2n+1})$$

U = objective displacement

$\therefore U_{2n} - U_{2n+1}$ be displacement of vibrations b/w $2n$ and $2n+1$ atoms.



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Force on $2n^{\text{th}}$ atom due to $(2n+1)^{\text{th}}$ atom:-

$$F_2 = -\beta U$$

$$F_2 = -\beta (U_{2n} - U_{2n+1})$$

$(U_{2n} - U_{2n-1})$ be displacement of vibrations b/w $2n^{\text{th}}$ and $(2n+1)^{\text{th}}$.

Net force on $2n^{\text{th}}$ atom due to its nearest atoms is

$$F = F_1 + F_2$$

$$F = -\beta (U_{2n} - U_{2n-1}) + \{-\beta (U_{2n} - U_{2n+1})\}$$

$$F = \beta [-U_{2n} + U_{2n-1} - U_{2n} + U_{2n+1}]$$

$$F = \beta [U_{2n-1} - 2U_{2n} + U_{2n+1}] \quad \text{--- (i)}$$

$$\text{As } F = m \vec{a}$$

$$F = m \frac{dv}{dt} = m \frac{d^2x}{dt^2} \quad \therefore v = \frac{dx}{dt}$$

The displacement of $2n^{\text{th}}$ atom is

U_{2n} , and $m = M$

$$F = m \frac{d^2 U_{2n}}{dt^2} \quad \text{--- (ii)}$$

Putting F in (i)

$$m \frac{d^2 U_{2n}}{dt^2} = \beta [U_{2n-1} - 2U_{2n} + U_{2n+1}] \quad \text{--- (iii)}$$

This eq of motion

For M_2 :-

Force on $(2n+1)^{\text{th}}$ atom due to $2n^{\text{th}}$ atom:-

$$F_1 = -\beta U$$

$$F_1 = -\beta [U_{2n+1} - U_{2n}]$$

$\therefore (U_{2n+1} - U_{2n})$ be displacement of vibrations b/w $(2n+1)^{\text{th}}$ atom and $2n^{\text{th}}$ atom.

$$\times F_2 = -\beta (U_{2n+2} - U_{2n+1}) \times$$

Force on $(2n+1)^{\text{th}}$ atom due to $(2n+2)^{\text{th}}$ atom

$$F_2 = -\beta U$$

$$F_2 = -\beta (U_{2n+1} - U_{2n+2})$$

Net force on $(2n+1)^{\text{th}}$ atom due to its nearest neighbouring

$$F = F_1 + F_2$$

$$F = -\beta [U_{2n+1} - U_{2n}] - \beta [U_{2n+1} - U_{2n+2}]$$

$$= +\beta [-U_{2n+1} + U_{2n} - U_{2n+1} + U_{2n+2}]$$

$$F = \beta [U_{2n+2} + U_{2n} - 2U_{2n+1}]$$

$$\text{As } F = m \frac{d^2 r}{dt^2}$$

The displacement of $(2n+1)^{\text{th}}$ atom is U_{2n+1} and $m = M_2$

So $F = m_2 \frac{d^2 U_{2n+1}}{dt^2}$ Putting in (iii) F

$$M_2 \frac{d^2 (U_{2n+1})}{dt^2} = \beta [U_{2n+2} + U_{2n-2} - 2U_{2n+1}] \quad \text{--- (iv)}$$

Now we want to find value of $U_{2n}, U_{2n+1}, U_{2n-1}, U_{2n+2}$
The harmonic solution for $2n^{\text{th}}$ atom

from its mean position is $U_{2n} = A e^{i(\omega t - kx)}$ $x = \text{displacement from mean position}$
(a) vector $k = \text{wave vector}$

where A is Amplitude of vibration of atoms having masses M

As displacement of $2n^{\text{th}}$ atom is $2na$ So $x = 2na$

Putting in (a)

$$U_{2n} = A e^{i(\omega t - k(2na))} \quad \text{--- (v)}$$

For $(2n+2)^{\text{th}}$, similarly

$$U_{2n+2} = A e^{i(\omega t - k(2n+2)a)} \quad \text{--- (vi)}$$

For $(2n+1)^{\text{th}}$ atom

$$U_{2n+1} = B e^{i(\omega t - k(2n+1)a)} \quad \text{--- (vii)}$$

As m_1 and m_2 both are different so their amplitude of vibrations are also different

So B is amplitude of vibration of atoms having mass M .
For $(2n-1)^{th}$ atom.

$$U_{2n-1} = B e^{i(\omega t - k(2n-1)a)} \quad \text{--- (viii)}$$

Now putting $U_{2n-1} = U_{2n} = U_{2n+1}$ in eq. (iii)

So

$$M \frac{d^2}{dt^2} [A e^{i(\omega t - k2na)}] = B [B e^{i(\omega t - k(2n-1)a)} - 2A e^{i(\omega t - k2na)} + B e^{i(\omega t - k(2n+1)a)}] \quad \text{--- (ix)}$$

consider

$$\frac{d^2}{dt^2} (A e^{i(\omega t - k2na)}) = \frac{d}{dt} (A e^{i(\omega t - k2na)}) \quad \text{--- (x)}$$

$$\frac{d^2}{dt^2} (A e^{i(\omega t - k2na)}) = A i^2 \omega^2 e^{i(\omega t - k2na)}$$

Putting in (ix)

$$M [A (i\omega)^2 e^{i(\omega t - k2na)}] = e^{i(\omega t - k2na)} [B e^{-ika} - 2A + B e^{ika}]$$

$$M A (-\omega^2) + 2AB = B B e^{-ika} + B B e^{ika}$$

$$2AB - M A \omega^2 = B B (e^{ika} + e^{-ika})$$

on R.H.S multiply and divided by 2

$$2AB - M A \omega^2 = B B 2 / (e^{ika} + e^{-ika})$$

$$2AB - M_1 \omega^2 A = 2\beta B \cos ka \quad \therefore \frac{e^{ika} + e^{-ika}}{2} = \cos ka$$

$$2A(2\beta - M_1 \omega^2) - 2\beta B \cos ka = 0 \quad \text{--- (X)}$$

Now Putting U_{2n+2} , U_{2n} , U_{2n+1} in (iv)

$$M_2 \frac{d^2}{dt^2} (B e^{i(\omega t - k_2 n a - k a)}) = \beta [A e^{i(\omega t - 2k_2 n a - 2ka)}]$$

$$+ A e^{i(\omega t - k_2 n a)} - 2B e^{i(\omega t - k_2 n a - ka)}] \quad \text{--- (Xi)}$$

Consider

$$\frac{d^2}{dt^2} (B e^{i(\omega t - k_2 n a - ka)}) = \beta i \omega e^{i(\omega t - k_2 n a - ka)}$$

$$\frac{d^2}{dt^2} (B e^{i(\omega t - k_2 n a - ka)}) = B(i\omega)^2 e^{i(\omega t - k_2 n a - ka)}$$

Putting this in (xi)

$$M_2 B(i\omega)^2 e^{i(\omega t - k_2 n a - ka)} = \beta [e^{i(\omega t - k_2 n a - ka)} [A e^{iak} + A e^{-ika} - 2B]]$$

$$-M_2 B \omega^2 = \beta [A e^{-iak} + A e^{ika}] - 2\beta B$$

$$-M_2 B \omega^2 + 2\beta B = A\beta \left[\frac{e^{ika} + e^{-ika}}{2} \right] \cdot 2$$

x and ÷ by 2 on RHS

$$B(2\beta - M_2 \omega^2) = 2A\beta [\cos ka]$$

$$B(2\beta - M_2 \omega^2) - 2A\beta [\cos ka] \quad \text{--- (Xii)}$$

← vibrations of 1
dimensional di-atomi...

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eq (i) and (ii) are homogenous
eqs in the unknowns A and B
A non-trivial solution exist
if determinant of co-efficients
are zero.

$$\begin{vmatrix} 2\beta - M_1\omega^2 & -2\beta \cos ka \\ -2\beta \cos ka & 2\beta - M_2\omega^2 \end{vmatrix} = 0$$

$\begin{vmatrix} A & B \\ C & D \end{vmatrix} = 0 \Rightarrow \begin{cases} Ax + By = 0 \\ Cx + Dy = 0 \end{cases}$
 $x \neq 0, y \neq 0$

$$(2\beta - M_1\omega^2)(2\beta - M_2\omega^2) - 4\beta^2 \cos^2 ka = 0$$

$$4\beta^2 - 2\beta M_2\omega^2 - 2\beta M_1\omega^2 + M_1M_2\omega^4 - 4\beta^2 \cos^2 ka = 0$$

$$M_1M_2\omega^4 - 2\beta\omega^2(M_1 + M_2) + 4\beta^2(1 - \cos^2 ka) = 0$$

$$M_1M_2\omega^4 - 2\beta\omega^2(M_1 + M_2) + 4\beta^2 \sin^2 ka = 0$$

$$\therefore 1 - \cos^2 ka = \sin^2 ka$$

$$M_1M_2 \left[\omega^4 - 2\beta\omega^2 \left(\frac{M_1 + M_2}{M_1M_2} \right) + 4\beta^2 \frac{\sin^2 ka}{M_1M_2} \right] = 0$$

Let $\omega^2 = \eta \Rightarrow \omega^4 = \eta^2$

$$\eta^2 - 2\beta \eta \left[\frac{M_1 + M_2}{M_1M_2} \right] + 4\beta^2 \left[\frac{\sin^2 ka}{M_1M_2} \right] = 0$$

To solve this eq using quadratic
formula. $ax^2 + bx + c = 0$.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

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$$\eta = \pm \frac{2\beta}{M_1 M_2} \left(\frac{M_1 + M_2}{2} \right) \pm \sqrt{\frac{4\beta^2}{M_1 M_2} \left(\frac{M_1 + M_2}{2} \right)^2 - \frac{4(1) \sin^2 ka}{M_1 M_2}}$$

$$\eta = \frac{2\beta}{M_1 M_2} \left(\frac{M_1 + M_2}{2} \right) \pm \sqrt{\frac{4\beta^2}{M_1 M_2} \left(\frac{M_1 + M_2}{2} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2}}$$

$$\eta = \frac{\beta}{M_1 M_2} \left(\frac{M_1 + M_2}{2} \right) \pm \sqrt{\frac{\beta^2}{M_1 M_2} \left(\frac{M_1 + M_2}{2} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2}}$$

$\therefore \eta = \omega^2$

$$\omega^2 = \frac{\beta}{M_1 M_2} \left(\frac{1}{M_2} + \frac{1}{M_1} \right) \pm \sqrt{\frac{\beta^2}{M_1 M_2} \left(\frac{1}{M_2} + \frac{1}{M_1} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2}}$$

This relation in terms of ω & k is called dispersion relation case of diatomic lattice.

For 3D diatomic lattice we get 2 values of ω for single value of k .

$$\omega_+^2 = \frac{\beta}{M_1 M_2} \left(\frac{1}{M_1} + \frac{1}{M_2} \right) + \sqrt{\frac{\beta^2}{M_1 M_2} \left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2}}$$

$$\omega_-^2 = \frac{\beta}{M_1 M_2} \left(\frac{1}{M_1} + \frac{1}{M_2} \right) - \sqrt{\frac{\beta^2}{M_1 M_2} \left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2}}$$

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Now we will consider different cases, for w_+ and w_- .

Case I: $\sin^2 ka \rightarrow 0$ then $\sin ka \rightarrow 0$

So w_+ will become

$$w_+^2 = \beta \left(\frac{1}{m_1} + \frac{1}{m_2} \right) + \beta \sqrt{\left(\frac{1}{m_1} + \frac{1}{m_2} \right)^2}$$

$$= \beta \left[\frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_1} + \frac{1}{m_2} \right]$$

$$w_+^2 = 2\beta \left[\frac{1}{m_1} + \frac{1}{m_2} \right]$$

$$w_+^2 = 2\beta \left[\frac{1}{m_1} + \frac{1}{m_2} \right]$$

$$w_+ = \sqrt{2\beta \left[\frac{1}{m_1} + \frac{1}{m_2} \right]}$$

angular frequency can't be -ive.

it is possible w_+ maximum value.

Case II $\sin^2 ka = 1$, $k = -\pi/2a$

$k = \pi/2a$ Putting in w_+

$$w_+^2 = \beta \left[\frac{1}{m_1} + \frac{1}{m_2} \right] + \sqrt{\left(\frac{1}{m_1} + \frac{1}{m_2} \right)^2 - \frac{4}{m_1 m_2}}$$

$$= \beta \left[\frac{1}{m_1} + \frac{1}{m_2} \right] + \sqrt{\left(\frac{1}{m_1} \right)^2 + \left(\frac{1}{m_2} \right)^2 + \frac{2}{m_1 m_2} - \frac{4}{m_1 m_2}}$$



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$$\omega_+^2 = \beta \left[\frac{1}{M_1} + \frac{1}{M_2} \right] + \beta \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2}}$$

$$\omega_+^2 = \beta \left[\frac{1}{M_1} + \frac{1}{M_2} \right] + \beta \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2}}$$

$$= \beta \left[\frac{1}{M_1} + \frac{1}{M_2} \right] + \beta \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2}}$$

$$\omega_+^2 = \beta \frac{2}{M_1}$$

$$\omega_+ = \sqrt{\frac{2\beta}{M_1}} \quad \text{Minimum value}$$

Case - III

$\sin^2 ka = 1, k = \frac{\pi}{2a}$

Putting ω_-^2

$$\omega_-^2 = \beta \left[\frac{1}{M_1} + \frac{1}{M_2} \right] - \beta \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2}}$$

$$= \beta \left[\frac{1}{M_1} + \frac{1}{M_2} \right] - \beta \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2}}$$

$$= \beta \left[\frac{1}{M_1} + \frac{1}{M_2} \right] - \beta \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4}{M_1 M_2}}$$

$$= \beta \left[\frac{1}{M_1} + \frac{1}{M_2} - \frac{1}{M_1} + \frac{1}{M_2} \right]$$

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$$\omega^2 = \frac{2\beta}{M_2} \quad , \quad \omega = \sqrt{\frac{2\beta}{M_2}}$$

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Chapter No:3

Question no 3 part (b)

Dispersion Curve

As we discussed above
diatomic linear lattice:

$$\omega_+^2 = \beta \left[\frac{1}{m} + \frac{1}{M} \right] + \beta \sqrt{\left[\frac{1}{m} + \frac{1}{M} \right] - \frac{4 \sin^2 ka}{mM}}$$

$$\omega_-^2 = \beta \left[\frac{1}{m} + \frac{1}{M} \right] - \beta \sqrt{\left[\frac{1}{m} + \frac{1}{M} \right] - \frac{4 \sin^2 ka}{mM}}$$

From Cases:

$$\omega_+ = \sqrt{2\beta \left(\frac{1}{m} + \frac{1}{M} \right)} \quad \text{at } k=0 \text{ (Max)}$$

$$\omega_+ = \sqrt{\frac{2\beta}{m}} \quad \text{at } k = \pm \frac{\pi}{2a} \text{ (Min)}$$

$$\omega_- = \frac{k_a}{\sqrt{mM}} \sqrt{2\beta}$$

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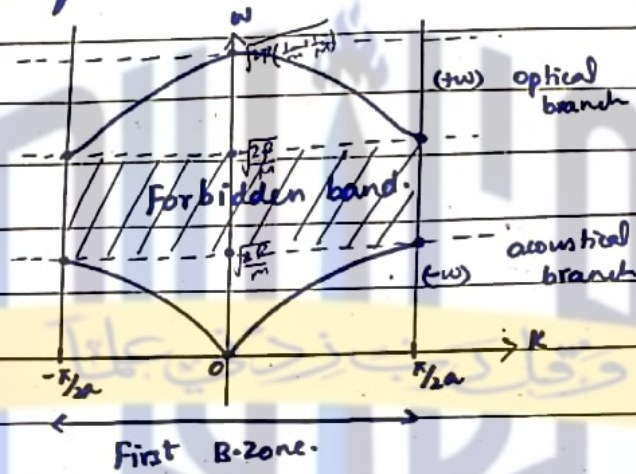


$$\sqrt{mM}$$

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$$\omega = 0 \text{ at } k=0 \quad (\text{Min})$$

$$\omega = \sqrt{\frac{2\beta}{M}} \text{ at } k = \pm \frac{\pi}{2a} \quad (\text{Max})$$

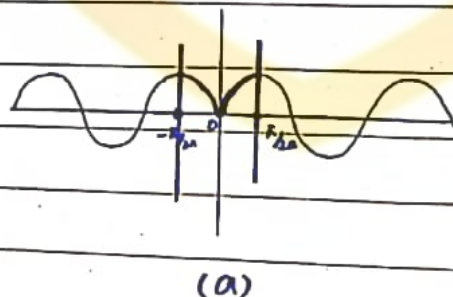
Graph between ω and k 

$$\because m < M$$

$$\omega^2 = \beta \left(\frac{1}{m} + \frac{1}{M} \right) \pm \beta \sqrt{\left(\frac{1}{m} + \frac{1}{M} \right)^2 - \frac{4 \sin^2 ka}{mM}}$$

Graph of $\sin^2 k$:-

(a)



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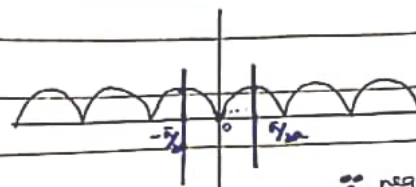




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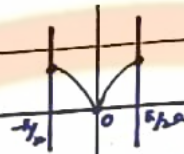
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(b)

negative portion
vanishes due to
 $\sin^2 k$.



(c)

(w+)

**Points:**

As we can see above figure,
we can get two branches;
upper branch known as
optical branch and lower
branch known as acoustical
branch.

(1) Range of Upper Branch

$$\sqrt{\frac{2\beta}{m}} < w < \sqrt{2\beta \left(\frac{1}{m} + \frac{1}{M} \right)}$$

(optical branch).

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$$\sqrt{\frac{2\beta}{m}} \quad \sqrt{\frac{2\beta}{M}}$$

(optical branch).

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(2) Range of Lower Branch

$$0 < \omega < \sqrt{\frac{2\beta}{m}}$$

(acoustical branch).

(3) Here we are getting a band gap OR frequency band in between optical branch and acoustical branch where frequency are not allowed in vibration of lattice. This is known as "Forbidden band."

Range of Forbidden band:-

$$\sqrt{\frac{2\beta}{M}} < \omega < \sqrt{\frac{2\beta}{m}}$$

Width of Forbidden band:-

$$\text{If } M \uparrow \Rightarrow \sqrt{\frac{2\beta}{M}} \downarrow$$

$$\text{If } m \downarrow \Rightarrow \sqrt{\frac{2\beta}{m}} \uparrow$$

$$(\text{width})_{F.B} \propto M$$

(width)



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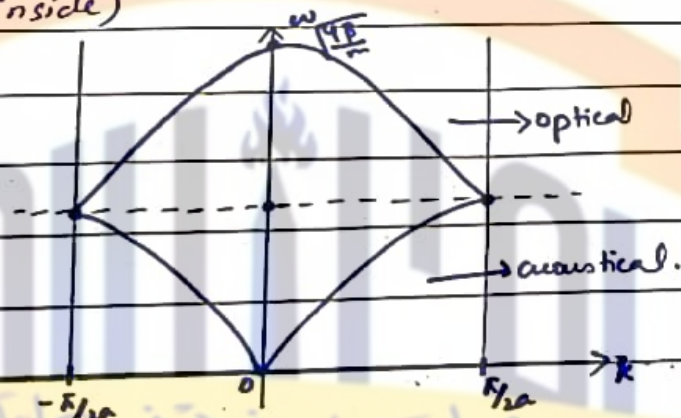
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$$\boxed{\text{(width)}_{F=13} = \frac{M}{m}}$$

(f) If $m = M$

$$\sqrt{\frac{2\beta}{m}} = \sqrt{\frac{2\beta}{M}}$$

(Coincide)



- For $m = M$
maxima of acoustical branch
coincide with minima of
optical branch at point $k = \pm \frac{\pi}{2a}$.

- For $m = M$

Forbidden band is disappears.

(5-) Range of First Brillouin Zone:-

$$-\frac{\pi}{2a} < k < \frac{\pi}{2a}$$

Smallest wavelength in this
zone;



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forbidden band
(5-) Range of First Brillouin Zone:-

$$-\frac{\pi}{2a} < k < +\frac{\pi}{2a}$$

Smallest wavelength in this zone;

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$$\lambda = \frac{2\pi}{k}$$

$$\lambda_{\min} = \frac{2\pi}{k_{\max}}$$

$$\lambda_{\min} = \frac{2\pi}{\pi/2a}$$

$$\lambda_{\min} = 4a$$

Smallest wavelength:

6

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Acoustic Branch

Dispersion relation according to acoustic branch is given by;

$$\omega^2 = \beta \left(\frac{1}{m} + \frac{1}{M} \right) - \beta \sqrt{\left(\frac{1}{m} + \frac{1}{M} \right)^2 - \frac{4 \sin^2 ka}{Mm}}$$

$$m = M_1, \quad M = M_2$$

$$\omega^2 = \beta \left(\frac{1}{M_1} + \frac{1}{M_2} \right) - \beta \sqrt{\left(\frac{1}{M_1} + \frac{1}{M_2} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2}}$$

$$\omega^2 = \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) - \beta \sqrt{\left(\frac{M_1 + M_2}{M_1 M_2} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2}}$$

$$\omega^2 = \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) - \beta \left[\left(\frac{M_1 + M_2}{M_1 M_2} \right)^2 - \frac{4 \sin^2 ka}{M_1 M_2} \right]^{1/2}$$

$$\omega^2 = \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) - \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right)^2 \left[1 - \frac{4 \sin^2 ka (M_1 M_2)^2}{(M_1 + M_2)^2} \right]$$

$$\omega^2 = \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) - \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) \sqrt{1 - \frac{4 \sin^2 ka (M_1 M_2)}{(M_1 + M_2)^2}}$$

Since,

acoustic branch begins at $k=0$

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Mathematically,

$$\lim_{k \rightarrow 0} (w_-) \rightarrow 0$$

So,

$$\sin^2 ka \approx k^2 a^2$$

∵ if $\theta \rightarrow 0$

$$\sin \theta \approx \theta$$

$$= \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) - \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) \left[\frac{1 - 4k^2 a^2 \frac{M_1 M_2}{(M_1 + M_2)^2}}{(M_1 + M_2)^2} \right]$$

$$= \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) - \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) \left[\frac{1 - 4k^2 a^2 \frac{M_1 M_2}{(M_1 + M_2)^2}}{(M_1 + M_2)^2} \right]^{1/2}$$

Take

$$\left[\frac{1 - 4k^2 a^2 \frac{M_1 M_2}{(M_1 + M_2)^2}}{(M_1 + M_2)^2} \right]^{1/2}$$

By applying binomial expansion
and neglecting higher term:

$$(a+b)^n = a^n + n a^{n-1} b + \dots$$

$$= 1 + \frac{1}{2} (1)^{-1/2} \cdot \left[\frac{-4k^2 a^2 \frac{M_1 M_2}{(M_1 + M_2)^2}}{(M_1 + M_2)^2} \right] + \dots$$

↓ neglect

$$= \left[1 - \frac{1}{2} \times 4k^2 a^2 \frac{M_1 M_2}{(M_1 + M_2)^2} \right]$$

$$= \left[1 - 2k^2 a^2 \frac{M_1 M_2}{(M_1 + M_2)^2} \right]$$

$$= \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) - \beta \left(\frac{M_1 + M_2}{M_1 M_2} \right) \left[\frac{1 - 2k^2 a^2 \frac{M_1 M_2}{(M_1 + M_2)^2}}{(M_1 + M_2)^2} \right]$$

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$$= \beta \frac{(M_1 + M_2)}{M_1 M_2} - \beta \frac{(M_1 + M_2)}{M_1 M_2} + 2\beta k^2 a^2 \frac{(M_1 + M_2)}{(M_1 M_2)^2}$$

$$= 2\beta k^2 a^2 \frac{1}{(M_1 + M_2)}$$

$$\omega^2 = \frac{2\beta k^2 a^2}{M_1 + M_2}$$

$$\omega^2 = \frac{2\beta k^2 a^2}{\sqrt{(M_1 + M_2)}}$$

$$\omega^2 = ka \left(\frac{2\beta}{M_1 + M_2} \right)$$

Phase Velocity:

$$\lim_{k \rightarrow 0} (V_p) = \lim_{k \rightarrow 0} \left(\frac{\omega}{k} \right)$$

$$V_p = a \sqrt{\frac{2\beta}{M_1 + M_2}}$$

Group Velocity:

$$\lim_{k \rightarrow 0} (V_g) = \lim_{k \rightarrow 0} \left(\frac{d\omega}{dk} \right)$$

$$V_g = a \sqrt{\frac{2\beta}{M_1 + M_2}}$$

Result:

As value of k increases, acoustic branch of curve rises first and then slowly decreases. Maximum value of frequency is;

$$(\omega)_{\max} = \sqrt{\frac{2\beta}{M_1}}$$

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$$(w)_{\max} = \sqrt{\frac{\gamma}{M_1}}$$

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Practical Displacement in acoustic branch:

Acoustical Branch:

Physical Meaning:-

In this branch, displacement of an atom is in same direction and amplitude of vibration is inversely proportional to the mass of atom.

⇒ If A is mass of M_1 and B is mass of M_2 .

Then,

$$A \propto \frac{1}{M_1}, \quad B \propto \frac{1}{M_2}$$

$$\frac{A}{B} = \frac{M_2}{M_1}$$

Take equation (13) and (14) from diatomic lattice.

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From eq (13)

$$A[M_1\omega^2 - 2\beta] + \beta^2 B \cos ka = 0$$



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Take equation (13) and (14)
from diatomic lattice.

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From eq. (13)

$$A[M_1\omega^2 - 2\beta] + \beta^2 B \cos ka = 0$$

When $(k \rightarrow 0)$

$$\cos ka = 1 - \frac{k^2 a^2}{2}$$

Put in eq. (13)

$$A[M_1\omega^2 - 2\beta] + \beta^2 B \left(1 - \frac{k^2 a^2}{2}\right) = 0$$

$$M_1\omega^2 A - 2\beta A + \beta^2 B - \beta k^2 a^2 B = 0 \rightarrow (16)$$

From eq. (14)

$$B(M_2\omega^2 - 2\beta) + \beta^2 A \cos ka = 0$$

When $(k \rightarrow 0)$

$$\cos ka = 1 - \frac{k^2 a^2}{2}$$

Put in eq. (14)

$$B(M_2\omega^2 - 2\beta) + \beta^2 A \left(1 - \frac{k^2 a^2}{2}\right) = 0$$

$$M_2\omega^2 B - 2\beta B + \beta^2 A - \beta k^2 a^2 A = 0 \rightarrow (17)$$

Add eq. (16) and (17)

$$M_1\omega^2 A - 2\beta A + 2\beta B - \beta k^2 a^2 B + M_2\omega^2 B - 2\beta B + \beta^2 A - \beta k^2 a^2 A = 0$$

$$M_1\omega^2 A - \beta k^2 a^2 A - \beta k^2 a^2 B + M_2\omega^2 B = 0$$

$$A(M_1\omega^2 - \beta k^2 a^2) - (\beta k^2 a^2 - M_2\omega^2)B = 0$$

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$$A(M_1\omega^2 - \beta k^2 a^2) - (\beta k^2 a^2 - M_2\omega^2)B = 0$$

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$$\frac{A}{B} = \frac{\beta k^2 a^2 - M_1\omega^2}{M_1\omega^2 - \beta k^2 a^2} \rightarrow (18)$$

For acoustical branch;

$$\omega^2 = \left(\frac{2\beta}{M_1 + M_2} \right) k^2 a^2$$

Put in eq (18)

$$\frac{A}{B} = \frac{\beta k^2 a^2 - M_2 \left[\left(\frac{2\beta}{M_1 + M_2} \right) k^2 a^2 \right]}{M_1 \left[\left(\frac{2\beta}{M_1 + M_2} \right) k^2 a^2 \right] - \beta k^2 a^2}$$

$$= \frac{\beta k^2 a^2 (M_1 + M_2) - 2\beta k^2 a^2 M_2}{(M_1 + M_2)}$$

$$= \frac{2\beta k^2 a^2 M_1 - \beta k^2 a^2 (M_1 + M_2)}{(M_1 + M_2)}$$

$$= \frac{\beta k^2 a^2 (M_1 + M_2) - 2\beta k^2 a^2 M_2}{2\beta k^2 a^2 M_1 - \beta k^2 a^2 (M_1 + M_2)}$$

$$= \frac{\beta k^2 a^2 M_1 + \beta k^2 a^2 M_2 - 2\beta k^2 a^2 M_2}{2\beta k^2 a^2 M_1 - \beta k^2 a^2 M_1 - \beta k^2 a^2 M_2}$$

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$$= \frac{\beta k a^+ M_1 + \beta k a^- M_1 - 2\beta k a^+ M_2}{2\beta k a^+ M_1 - \beta k a^+ M_1 - \beta k a^- M_2}$$

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$$= \beta k a^+ M_1 - \beta k a^- M_2$$

$$\beta k a^+ M_1 - \beta k a^- M_1$$

$$\frac{A}{B} = 1$$

Hence,

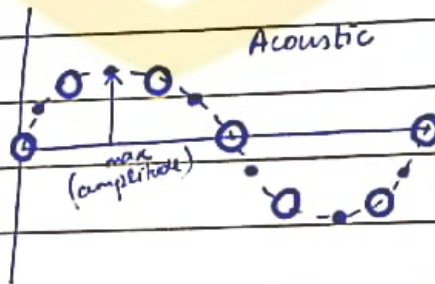
the two masses will move in same direction with some amplitude. The atoms move together in long wavelength acoustical vibrations.

In Equilibrium:-

• O • O • O • O • O • O (without disturbance)



Then;

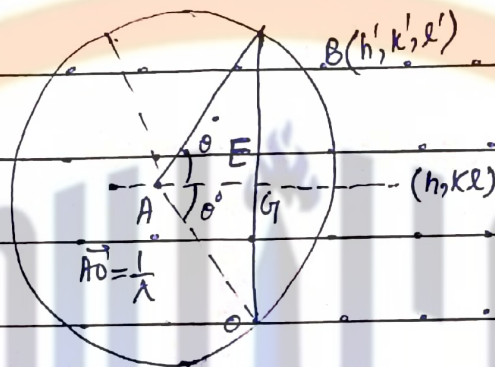


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Q No: 4 (a)

Derive the diffraction condition in reciprocal lattice.

Consider a reciprocal lattice space which has a large No of reciprocal lattice points as shown in fig.



$$AO = \frac{1}{\lambda}$$

Construction steps:-

- 1) A vector $\vec{AO}$ of length $\frac{1}{\lambda}$ is drawn in the direction of incident X-ray beam. The point 'A' is not necessarily a lattice point of reciprocal lattice and point 'O' is the origin of reciprocal lattice.
- 2) Draw a circle of radius $AO = \frac{1}{\lambda}$ taking A as centre. This circle is known as Ewald's circle in 2D space and as Ewald's sphere in 3D space.
- 3) Let the circle intersect some point (h', k', l') of the reciprocal lattice at B .
- 4) $\vec{OB}$ represents a vector connecting the origin 'O' of

the reciprocal lattice and a point $B(h', k', l')$ of the reciprocal lattice.

$$\vec{OB} = \vec{G} = \text{Reciprocal lattice vector}$$

- 5) Draw perpendicular bisector AE of OB . Thus AE represents (hkl) plane of direct lattice.

$$|\vec{OB}| = |\vec{G}| = \frac{1}{d_{hkl}} \quad \text{--- (1)}$$

where d is interplanar spacing in (hkl) plane of direct lattice

In $\triangle OAE$:-

$$\sin \theta = \frac{OE}{AO} \Rightarrow \sin \theta = \frac{OE}{\lambda}$$

$$OE = \frac{\lambda \sin \theta}{2}$$

$$\text{So } |\vec{OB}| = 2OE$$

$$|\vec{OB}| = 2 \times \frac{\lambda \sin \theta}{2} \rightarrow \text{--- (2)}$$

Compare (1) and (2)

$$\frac{1}{d} = \frac{2 \sin \theta}{\lambda}$$

$$\lambda = 2d \sin \theta \quad \text{For first order } n=1$$

$$\text{In general law } 2d \sin \theta = n\lambda$$

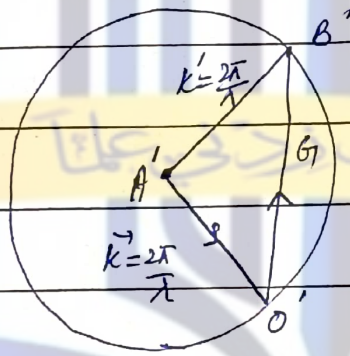
Vector form of Bragg's law OR

Diffraction condition in Reciprocal lattice:-

To convert $\vec{AO}$ into wave vector $\vec{k}$, it is multiplied by 2π .

$$\vec{AO} = \frac{1}{\lambda} = 2\pi \times \frac{1}{\lambda} = \frac{2\pi}{\lambda} = \vec{k}$$

For this $\vec{AB}$ vector is also multiplied by 2π , so radius of circle will increase 2π times.



$\vec{A'O'}$ is 2π times greater than $\vec{AO}$.

With the help of Ewald's construction, we can write Bragg's law in vector form. Let all the vectors drawn in fig 0 are magnified by multiple of 2π .

$\vec{A'O'} = \vec{k}$ = Incident wave vector

$$|\vec{k}| = \frac{2\pi}{\lambda}$$

$\vec{A'B'} = \vec{k'}$ = Diffracted wave vector

$$|\vec{k'}| = \frac{2\pi}{\lambda}$$

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Using law of vector addition

$$\vec{A'O'} + \vec{O'B'} = \vec{AB'}$$

$$\vec{K} + \vec{G} = \vec{K'}$$

$$\text{Since } \vec{A'O'} = \vec{AB'}$$

$$\text{So } \vec{K} = \vec{K'}$$

$$\vec{K} = \vec{G} + \vec{K'}$$

$$(\vec{K})^2 = (\vec{K} + \vec{G})^2$$

$$K^2 = K^2 + G^2 + 2KG$$

$$\boxed{(\vec{G})^2 + 2\vec{K}\vec{G} = 0}$$

• Madelung Energy: ques no 4 part (b)

"Madelung energy is the total potential energy of an ionic crystal lattice due to the electrostatic interactions b/w the ions within the crystal structure."

"OR"

"The long range interaction b/w ions with charge $\pm q$ is the electrostatic interaction $\pm \frac{q^2}{r}$, attractive b/w ions of opposite charge and repulsive b/w ions of the same charges. The main contribution to the binding energy of ionic crystals is electrostatic and is called Madelung energy."

Calculations:

The sum of all interaction for the i^{th} atom can be written as: $U_i = \sum' U_{ij} \quad \text{--- (1)}$

where the primed summation includes all ions except $j=i$

$$U_{ij} = \left(\exp(-\lambda_{ij}/\rho) \pm \frac{q^2}{\epsilon_{ij}} \right) \quad \text{--- (2)}$$

where $\pm$ represents interactions b/w like and opposite charges.

$\exp(-\lambda_{ij}/\rho)$ = central field repulsive potential
 λ, ρ are empirical parameters.

It is convenient to introduce dimensionless parameter.

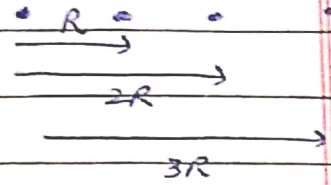
$$g_{ij} = p_{ij} R$$

where R is the nearest neighbour separation in the crystal.

$$\text{As } g_{ij} = p_{ij} R:$$

The eq (2) will become:

$$U_{ij} = \exp\left(\frac{-p_{ij} R}{\rho}\right) \pm \frac{q^2}{p_{ij} R}$$



$p_{ij} = 1$ for repulsive interaction. As it is very short range for first neighbour ion. so;

$$U_{ij} = \exp\left(\frac{-R}{\rho}\right) \pm \frac{q^2}{p_{ij} R} \quad \text{--- (3)}$$

The total lattice energy U_{total} of a crystal is composed of N molecules.

$$U_{\text{total}} = N U_i$$

So eq no (3) becomes:-

$$U_{\text{total}} = N U_i = N z \left[\exp\left(\frac{-R}{\rho}\right) \pm \sum_j \frac{q^2}{p_{ij} R} \right] \quad \text{--- (4)}$$

where z is the number of nearest neighbours of any ions.

• Madelung Constant:

From eq no (4):-

$$\alpha = \sum_j \pm \frac{1}{p_{ij}}$$

where (+) sign is taken for like charges and (-) sign for unlike charges.

The value of the Madelung constant is of central importance in the theory of ionic crystals.

Eq no (4) becomes:-

$$U_{\text{total}} = N \left[z \left[\exp\left(\frac{-R}{\rho}\right) - \alpha \frac{q^2}{R} \right] \right] \quad \text{--- (5)}$$

• At Equilibrium Separation:-

$$\left(\frac{dU_{\text{total}}}{dR} \right)_{R=R_0} = 0$$

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$$-2\lambda e^{-R/\rho} + \alpha q^2 = 0$$

$$\frac{\alpha q^2}{R_0^2} = \frac{2\lambda e^{-k/\rho}}{\rho}$$

$$4ze^{-R/\rho} = \sigma \eta^2 \rho / R_0^2 \quad \text{--- (6)}$$

$$\rho_0 e^{-\frac{R_0}{r}} = \frac{\rho_0 q^2}{21}$$

Put eq no (6) in eq (5) :-

~~$$H_{\text{blue}} = N \left(\frac{\alpha q^2 \rho}{R_0^2} \right) - \frac{\alpha q^2}{R_0}$$~~

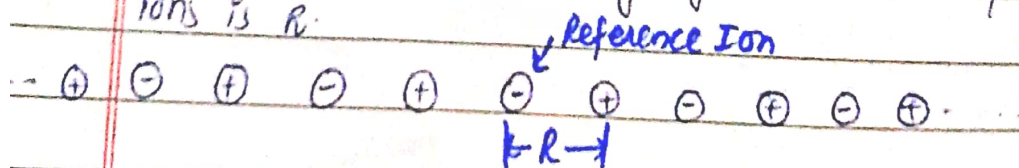
$$U_{tot} = -N \alpha \frac{q^2}{R_0} \left(1 - \frac{\rho}{R_0}\right)$$

The Term $\frac{Nq^2}{R_0}$ is the Madelung Energy. The ρ can be determine of the order 0.1 R_0 . So, the repulsive attraction has a very short range.

- Evaluation of Madelung constant for an infinite one dimensional array of ions

$$\alpha = \frac{\sum \pm 1}{p_{ij}}$$

A line of ions of alternating sign with distance b/w ions is R .



By dividing R on b.s of eq (7)

$$\frac{\alpha}{R} = \sum_j \pm \frac{1}{r_{ij} R}$$

$$\therefore h_j = h_j k$$

$$\frac{\alpha}{R} = \sum_j \pm \frac{1}{R_{ij}}$$

$$\frac{\alpha}{R} = \left[\frac{1}{R} - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \dots + \frac{1}{R} - \frac{1}{2R} + \frac{1}{3R} - \frac{1}{4R} + \dots \right]$$

$$\frac{\alpha}{R} = \left[\frac{2}{R} - \frac{2}{2R} + \frac{2}{3R} - \frac{2}{4R} + \dots \right] \Rightarrow \frac{\alpha}{R} = \frac{2}{R} \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right]$$

$$\frac{\alpha}{R} = \frac{2}{R} \ln(1+1) \Rightarrow \frac{\alpha}{R} = \frac{2}{R} \ln 2 \Rightarrow \alpha = 2 \ln 2$$

$$\alpha = 2 \ln 2 \Rightarrow \boxed{\alpha = 1.386}$$

$$\because \ln(1+n) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

if $n=1$

$$\because \ln(1+1) = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

It is value of Madelung constant.

Values of Madelung constant for different

crystals are-

$$\text{NaCl} = \alpha = 1.7475$$

$$\text{CsCl} = \alpha = 1.7626$$

$$\text{ZnS} = \alpha = 1.6381$$

Einstein theory of specific heat

OR

Quantum theory of lattice heat capacity

Einstein in 1911 resolved the problem of classical theory of lattice heat capacity on the basis of Quantum theory.

Einstein retained all the assumptions of Classical theory except replacing the classical harmonic oscillators by quantum harmonic oscillators which can have only discrete energy values.

Salient features of Einstein Theory:

- *1: The atoms of crystals are identical and independent harmonic oscillators.

- *2:- A solid (contain)ⁿ consisting N atoms is equivalent to $3N$ one dimensional harmonic oscillators.
- *3 All the oscillators vibrate with the same natural frequency due to identical environment of each.
- *4 The oscillators are quantum oscillators and have discrete energy value rather than continuous as in classical theory.
- *5 Any number of oscillators may be present in the same quantum state.
- *6 The atomic oscillators are identicals and identifiable due to their location at separate lattice sites and hence obey the Maxwell-Boltzmann distribution of energies.

In classical model, the vibrational energy can change continuously. This ideal of quantized energy level was taken from the work of **Planck**, who for the first time introduced the idea of quantum of energy for explaining successfully the problem of black-body radiation.

$$E_n = nh\nu = n \frac{h}{2\pi} (2\pi\nu) = n\hbar\omega_0$$

$$\therefore 2\pi\nu = \omega_0$$

$$\therefore \frac{h}{2\pi} = \hbar$$

Where $n = 0, 1, 2, 3, \dots$ is called the quantum number.

Later, it was realized that it is better to express energy by using $E_n = (n + \frac{1}{2})\hbar\omega_0$ rather than $E_n = nh\nu$.

So:

Later, it was realized that
it is better to express energy
by using $E_n = (n + \frac{1}{2}) \hbar \omega_0$
rather than $E_n = n \hbar \omega_0$.

So;

Planck's Quantum theory, the discrete
energy values of an oscillator

$$E_n = (n + \frac{1}{2}) \hbar \omega_0 ; n = 0, 1, 2, 3, \dots$$

• $n = 0$

→ Eq (1)

$$E_0 = \frac{1}{2} \hbar \omega_0 \rightarrow \text{Zero point Energy}$$

• $n = 1$

$$E_1 = \frac{3}{2} \hbar \omega_0$$

• $n = 2$

$$E_2 = \frac{5}{2} \hbar \omega_0$$

⇒ **Note**

- In classical theory, when temp is zero $^{\circ}\text{K}$ then the energy of an oscillator is also zero.
- In Quantum theory of lattice Heat capacity. When temp is zero kelvin " $^{\circ}\text{K}$ " then the atom is vibrate to its mean position with the Zero point Energy. This statement shows that $E_0 = \frac{1}{2} \hbar \omega_0$ are Temperature independent at $^{\circ}\text{K}$.

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Now Using Maxwell Boltzmann distribution of energy by Replacing f by Σ because oscillators took quantized values and not continuously vary energy.

Following Steps for Derivation:

Step 1: Find the average thermal energy of single one dimensional S.H.O (Simple Harmonic Oscillator).

Step 2: Find the total energy

Step 3: Find the C_v Value.

Step 4: Special Cases.

Step 1:-

The average energy

$$\bar{E} = \frac{\sum_{n=0}^{\infty} E_n \exp\left(\frac{-E_n}{k_B T}\right)}{\sum_{n=0}^{\infty} \exp\left(\frac{-E_n}{k_B T}\right)} \quad \text{--- (2)}$$

The limit of summation has been taken as $0 \rightarrow \infty$, because the number of atoms in the

crystal is very large.

Using Eq (1) in Eq (2)

$$\bar{E} = \frac{\sum_{n=0}^{\infty} (n + \frac{1}{2}) \hbar \omega_0 \exp\left(\frac{-(n + \frac{1}{2}) \hbar \omega_0}{k_B T}\right)}{\sum_{n=0}^{\infty} \exp\left(\frac{-(n + \frac{1}{2}) \hbar \omega_0}{k_B T}\right)}$$

$$\left(= \sum_{n=0}^{\infty} \right) \bar{E} = \hbar \omega_0 \frac{\sum_{n=0}^{\infty} (n + \frac{1}{2}) \exp\left(\frac{-(n + \frac{1}{2}) \hbar \omega_0}{k_B T}\right)}{\sum_{n=0}^{\infty} \exp\left(\frac{-(n + \frac{1}{2}) \hbar \omega_0}{k_B T}\right)}$$

Putting value of $n=0, 1, 2, \dots$

$$\bar{E} = \hbar \omega_0 \left[\frac{1}{2} \exp\left(\frac{-\frac{1}{2} \hbar \omega_0}{k_B T}\right) + \frac{3}{2} \exp\left(\frac{-\frac{3}{2} \hbar \omega_0}{k_B T}\right) + \frac{5}{2} \exp\left(\frac{-\frac{5}{2} \hbar \omega_0}{k_B T}\right) + \dots \right]$$

$$\left[\exp\left(\frac{-\frac{1}{2} \hbar \omega_0}{k_B T}\right) + \exp\left(\frac{-\frac{3}{2} \hbar \omega_0}{k_B T}\right) + \exp\left(\frac{-\frac{5}{2} \hbar \omega_0}{k_B T}\right) + \dots \right]$$

• Let $x = \frac{-\hbar \omega_0}{k_B T}$

$$\bar{E} = \hbar \omega_0 \left[\frac{1}{2} e^{+\frac{1}{2}x} + \frac{3}{2} e^{\frac{3}{2}x} + \frac{5}{2} e^{\frac{5}{2}x} + \dots \right]$$

$$\left[e^{\frac{1}{2}x} + e^{\frac{3}{2}x} + e^{\frac{5}{2}x} + \dots \right]$$

$$\bar{E} = \hbar \omega_0 \frac{d}{dx} \ln \left[e^{\frac{1}{2}x} + e^{\frac{3}{2}x} + e^{\frac{5}{2}x} + \dots \right]$$

$$\bar{E} = \hbar \omega_0 \frac{d}{dx} \ln \left[e^{\frac{1}{2}x} (1 + e^x + e^{2x} + \dots) \right]$$

$$\bar{E} = \hbar \omega_0 \frac{d}{dx} \left[\ln e^{\frac{1}{2}x} + \ln(1 + e^x + e^{2x} + \dots) \right]$$

$$\bar{E} = k\omega_0 \frac{d}{dx} \left[\frac{x}{2} \ln e + \ln(1 + e^x + e^{2x} + \dots) \right]$$

• Sum of infinity series

$$\bullet \log e^n = n \ln e$$

$$G \cdot P = S_{\infty} = \frac{a}{1-r}$$

• Where a is 1st term

$$r = \frac{\text{2nd term}}{\text{1st term}} = \frac{e^x}{1} = e^x$$

$$\bar{E} = k\omega_0 \frac{d}{dx} \left[\frac{x}{2} + \ln \left(\frac{1}{1-e^x} \right) \right]$$

$$\bar{E} = k\omega_0 \left[\frac{1}{2} + \frac{d}{dx} \ln(1-e^x)^{-1} \right] \quad \therefore n \ln a$$

$$\bar{E} = k\omega_0 \left[\frac{1}{2} - \frac{d}{dx} (\ln(1-e^x)) \right]$$

$$\bar{E} = k\omega_0 \left[\frac{1}{2} - \frac{(-e^x)}{1-e^x} \right]$$

$$\bar{E} = k\omega_0 \left[\frac{1}{2} + \frac{e^x}{1-e^x} \right]$$

$$\bar{E} = k\omega_0 \left[\frac{1}{2} + \frac{e^x}{e^x(e^{-x}-1)} \right]$$

$$\bar{E} = k\omega_0 \left[\frac{1}{2} + \frac{1}{e^{-x}-1} \right]$$

$$\bar{E} = \frac{1}{2} k\omega_0 + \frac{k\omega_0}{e^{-x}-1}$$

- put $x = -\frac{\hbar\omega_0}{k_B T}$

$$\bar{E} = \frac{1}{2} \hbar\omega_0 + \frac{\hbar\omega_0}{e^{-(-\frac{\hbar\omega_0}{k_B T})} - 1}$$

$$\boxed{\bar{E} = \frac{1}{2} \hbar\omega_0 + \frac{\hbar\omega_0}{e^{\frac{\hbar\omega_0}{k_B T}} - 1}} \quad \text{--- ①}$$

At $T = 0\text{K}$

$$\bar{E} = \frac{1}{2} \hbar\omega_0 + \frac{\hbar\omega_0}{e^{\frac{\hbar\omega_0}{k_B(0)}} - 1}$$

$$\bar{E} = \frac{1}{2} \hbar\omega_0 + \frac{\hbar\omega_0}{e^\infty - 1}$$

$$\bar{E} = \frac{1}{2} \hbar\omega_0 + \frac{\hbar\omega_0}{\infty - 1} = \frac{1}{2} \hbar\omega_0 + \frac{\hbar\omega_0}{\infty} \rightarrow 0$$

$$\boxed{\bar{E} = \frac{1}{2} \hbar\omega_0} \rightarrow \text{Zero point Energy.}$$

This is also temp

independent term.

So, we can say that; average thermal energy of single one dimensional S.H.O.

Step 2:

Find total Energy.

$$E = 3N\bar{E}$$

$$E = 3N \left[\frac{1}{2} \hbar \omega + \frac{\hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} \right]$$

$$E = \frac{1}{2} 3N \hbar \omega + 3N \frac{\hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} \quad (4)$$

This is total energy of the $3N$ one dimensional harmonic oscillator.

Step 3:

Find the C_V value.

$$C_V = \left(\frac{\partial E}{\partial T} \right)_V$$

$$C_V = \frac{d}{dT} \left[\frac{1}{2} 3N \hbar \omega + 3N \frac{\hbar \omega}{e^{\frac{\hbar \omega}{k_B T}} - 1} \right]$$

$$C_V = 0 + 3N \hbar \omega \frac{d}{dT} \left[\left(e^{\frac{\hbar \omega}{k_B T}} - 1 \right)^{-1} \right]$$

$$C_V = 3N \hbar \omega (-1) \cdot \left(e^{\frac{\hbar \omega}{k_B T}} - 1 \right)^{-2} \cdot e^{\frac{\hbar \omega}{k_B T}} \cdot \left(-\frac{\hbar \omega}{k_B T^2} \right)$$

$$C_V = 3N \hbar \omega \left(\frac{\hbar \omega}{k_B T} \right)^2 \frac{e^{\frac{\hbar \omega}{k_B T}}}{\left(e^{\frac{\hbar \omega}{k_B T}} - 1 \right)^2}$$

Multiply & divide by k_B .

$$C_V = 3N k_B \left(\frac{\hbar \omega}{k_B T} \right)^2 \frac{e^{\frac{\hbar \omega}{k_B T}}}{\left(e^{\frac{\hbar \omega}{k_B T}} - 1 \right)^2} \quad (5)$$

This is the value of C_v .

$$\text{Let } \theta_E = \frac{h\nu_0}{k_B}$$

Where θ_E is the Einstein Temp or characteristic Temp.

Where

$$Nk_B = R$$

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \cdot \frac{e^{-\frac{\theta_E}{T}}}{\left(e^{-\frac{\theta_E}{T}} - 1 \right)^2} \quad (6)$$

Step 4:
Special Cases:

At high Temp:-

A high Temp $e^{\frac{\theta_E}{T}} = 1 + \frac{\theta_E}{T} + \dots$

→ By using $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

So, eq (6) become

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \cdot \frac{\left(1 + \frac{\theta_E}{T} \right)}{\left(1 + \frac{\theta_E}{T} + 1 \right)^2}$$

$$C_v = 3R \left(\frac{\theta_E}{T} \right)^2 \cdot \frac{\left(1 + \frac{\theta_E}{T} \right)}{\left(\frac{\theta_E}{T} \right)^2}$$

$$C_V = 3R \left(1 + \frac{\theta_E}{T} \right)$$

Since $\theta_E \ll T$

So,

$$\frac{\theta_E}{T} \rightarrow 0$$

$$C_V = 3R(1+0)$$

$$[C_V = 3R]$$

By this classical & Petit theory agree each other.

⇒ At Low Temp.

At low Temp $T \ll \theta_E$

$$(e^{\theta_E/T} - 1) \approx e^{\theta_E/T}$$

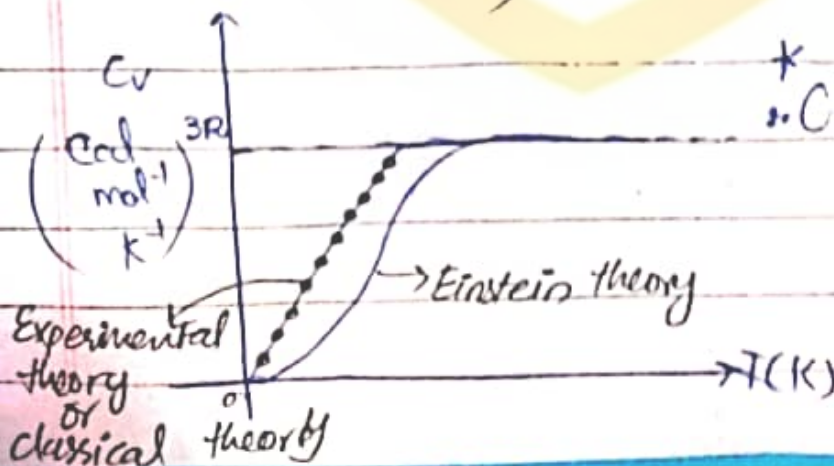
∴ (6) become

$$C_V = 3R \left(\frac{\theta_E}{T} \right)^2 \cdot \frac{\theta_E}{T} \cdot \frac{1}{(e^{\theta_E/T})^3}$$

$$C_V = 3R \left(\frac{\theta_E}{T} \right)^2 \cdot (e^{\theta_E/T})^{-1}$$

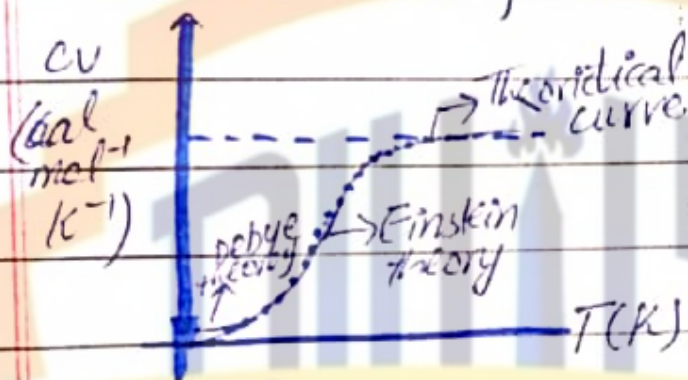
$$[C_V \propto e^{-\frac{\theta_E}{T}}] \text{ for } T \ll \theta_E, \text{ the}$$

heat capacity is proportional to $e^{-\theta_E/T}$. So,



→ Failure & Draw-back :

- At low Temp, C_v decrease with Temp exponentially which is different from experimental curve where $C_v \propto T^3$. (Debye's Law), which is draw-back of this Model.



- This model physically is unrealistic because all the vibration modes cannot have same frequency unless all the atom in the solid vibrate in a manner independent of each other.
- The movements of adjacent atoms always tends to be correlated because of the coupling force within the solid.